

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – MCQ Test

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 3 – MCQ Test
Weightage:	30% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read each question carefully before selecting your answer.
- This assessment consists of 50 multiple-choice questions.
- Do not use external resources, AI tools, textbooks, or assistance from others during the assessment unless explicitly permitted by the instructor.
- Upload Score Card in GCR and submit score card printout.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Context-Free Grammars (CFG)	CFG components, productions, terminals, non-terminals, recursion, sentence generation	1 – 10
Sentence Construction and Agreement	Parse trees, sentence structure, grammatical agreement, syntactic validation	11 – 20
Subcategorization	Verb frames, complement requirements, argument structure analysis	21 – 25
Top-Down Parsing	Parsing strategy, derivation process, recursion issues, grammar expansion	26 – 30
Earley Parsing	Predictor, Scanner, Completer operations, ambiguity handling	31 – 35
Feature Structures	Feature representation, unification, constraint checking, agreement enforcement	36 – 40
Probabilistic Context-Free Grammars (PCFG)	Rule probabilities, ambiguity resolution, probabilistic parsing	41 – 48
Integrated Syntax Analysis	CFG + Feature Structures + PCFG integration in NLP applications	49 – 50

Annexure A – MCQ Test

<https://forms.gle/WSv28zfaU5B2aWcy6>

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Number of Correct Answers	100%	45 to 50 correct answers	38 to 44 correct answers	25 to 37 correct answers	0 to 24 correct answers
Accuracy	100%	Excellent (90–100%)	Good (75–89%)	Average (50–74%)	Poor (Below 50%)

No. of Questions	Number of Correct Answers	Accuracy	Total %
50			

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Name: B. Akshaya

Regno: 192424936

DSA0304

Test - 2

- 1) B (recursive production rules allow hierarchical generation)
- 2) A (subject-verb agreement)
- 3) B (parse tree)
- 4) B
- 5) B
- 6) B
- 7) C
- 8) A
- 9) B
- 10) A
- 11) B
- 12) B
- 13) B
- 14) B
- 15) D
- 16) A
- 17) B
- 18) A

44) A

45) A

46) A

47) C

48) B

49) A

50) C

45
50

- 19) Earley parsing handling ambiguity efficiently, while PCFG standard
- 20) CFG $\rightarrow$ checking $\rightarrow$ feature unification $\rightarrow$ PCFG based parser (C)

- 21) C (Introduce a compound noun production N-'Livingroom')
- 22) A (Main subject noun phrase) ✓
- 23) Verb subcategorization frame (B) ✓
- 24) Shift-reduce conflict (A) ✓
- 25) Predictor (A) ✓
- 26) Maximum likelihood estimation (B) ✓
- 27) Tree A because it has the highest probability (A) ✓
- 28) Context-free Grammar only ✓
- 29) Recursive Top-down parser (B) ✓
- 30) Maximum Frequency Matching (A) ✓
- 31) (A) ✓
- 32) (D) ✓
Engineers develop the software
- 33) She gave the customer a refund (C) ✓
- 34) Build parse tree (C) ✓
- 35) Predictor ✓
- 36) (B) unification fails because feature values conflict ✓
- 37) (A) the parse with highest overall probability ✓
- 38) Recursive Descent parsing (B) ✓
- 39) (A) ✓
- 40) (B) ✓
- 41) (B) $Np \rightarrow Det + N$ ✓
- 42) Det (A) ✓
- 43) Complete (B) ✓